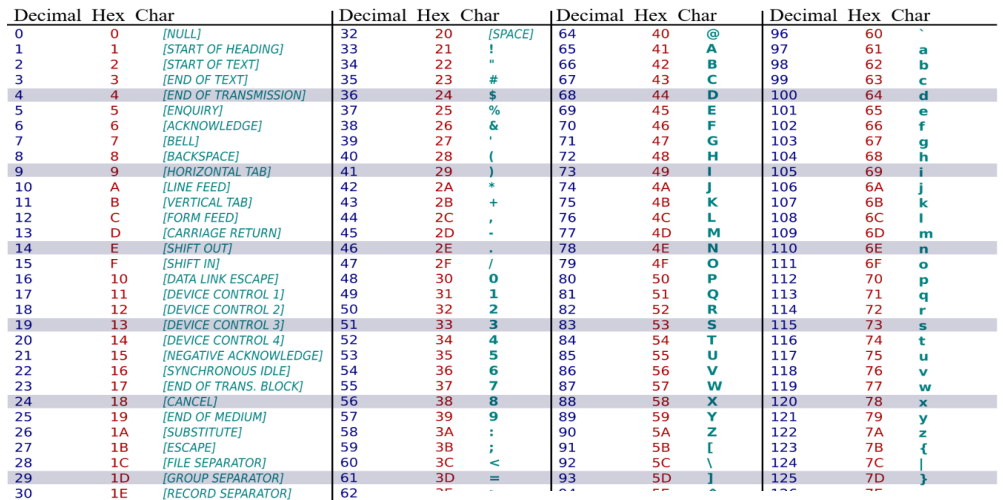


- Transmit the result back via computer's terminal and start over



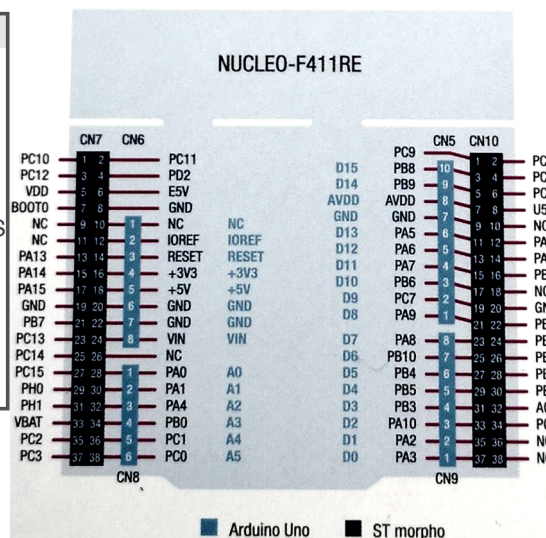
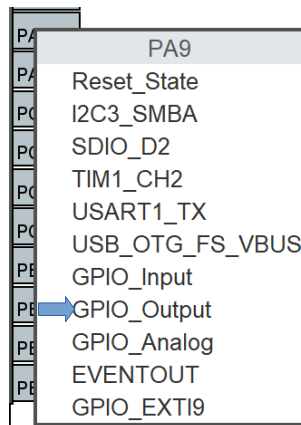
```
int state=0;
uint32_t i,j;
char c[3];

uint32_t lastPushTick =0;
while (1)
{
    if (HAL_UART_Receive(&huart2, c+3, 1, 100) == HAL_OK) {
        for(int i=0;i<3;i++){
            c[i]=c[i+1];
        }

        if(c[2]!='\n'){
            c[1]+=state;
            state=0;
            HAL_UART_Transmit(&huart2, c+1, 1, 100);
            HAL_UART_Transmit(&huart2, c+2, 1, 100);
            c[2]=0;
        }

        if (HAL_GPIO_ReadPin(GPIOC, GPIO_PIN_13) == GPIO_PIN_RESET) {
            lastPushTick=HAL_GetTick();
            state=(state+1);

            while(HAL_GetTick()-lastPushTick<=100)
            while(HAL_GPIO_ReadPin(GPIOC,GPIO_PIN_13)==GPIO_PIN_RESET) ;
        }
    }
}
```



```

    Bool pressed=0;
    /* USER CODE BEGIN PV */
    UART_HandleTypeDef huart2;
    /* USER CODE END PV */

    /* Private function prototypes -----
void SystemClock_Config(void);
static void MX_GPIO_Init(void);
static void MX_USART2_UART_Init(void)
    /* USER CODE BEGIN PFP */

void UART2_Init(void)
{
    int state=0;

    char c[]=(54,53,51,49,51,52,49,56,50,49);
    uint32_t lastPushTick =0;
    uint32_t lastPushTick2 =0;
    while (1)
    {
        if (HAL_GetTick()-lastPushTick>500) {
            HAL_GPIO_TogglePin(GPIOA,GPIO_PIN_5);
            lastPushTick = HAL_GetTick();

            if (HAL_GPIO_ReadPin(GPIOA,
                state=(state+1)%10;
                HAL_UART_Transmit(&huart2,
                    lastPushTick2 = HAL_GetTick();
                    while (HAL_GetTick()-lastPushTick2>500)
                    while (HAL_GPIO_ReadPin(GPIOA,GPIO_PIN_5)==0)
                    if (HAL_GetTick()-lastPushTick2>500)
                }
            }
        }
    }
}

```

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Skill Test Sec 1 2562

1. Make red and green lights blink every 150 ms.
2. Use user push button to "TOGGLE" to make red and green lights blink every 30 ms.
3. Hold push button at least 2 s. to stop blinking and make program doesn't response to push button anymore.

```
int state=0;
int c[]={30,150};

int lastPushTick=0;
int lastPushTick2=0;
/* USER CODE END 2 */

/* Infinite loop */
/* USER CODE BEGIN WHILE */
while (1)
{
    if (HAL_GetTick()-lastPushTick>=c[state]){
        HAL_GPIO_TogglePin(GPIOA,GPIO_PIN_5);
        HAL_GPIO_TogglePin(GPIOA,GPIO_PIN_9);
        lastPushTick = HAL_GetTick();

        if (HAL_GPIO_ReadPin(GPIOC, GPIO_PIN_13) == GPIO_PIN_RESET) {
            state=(state+1)%2;
            lastPushTick2 = HAL_GetTick();
            while (HAL_GetTick()-lastPushTick2<=100)
            while (HAL_GPIO_ReadPin(GPIOC,GPIO_PIN_13)==GPIO_PIN_RESET);
            if (HAL_GetTick()-lastPushTick2>=200){ while(!){HAL_GPIO_WritePin(GPIOC,GPIO_PIN_5,GPIO_PIN_RESET);}}
        }
    }
}
/* USER CODE END WHILE */
```

Skill Test Sec 2 2562

1. Make all lights blink every 150 ms (on 50 ms, off 100 ms)
2. Use user push button to "TOGGLE" to make all lights blink (on 100 ms, off 50) สลับกับข้อบน
3. Hold push button at least 2 s. to make all lights on and make program doesn't response to push button anymore.

```
int state=0;
char buf[256];
char buttoncountwrite[256];
int buttoncount=0;
uint32_t lastPushTick =0;

while (1)
{
    /* USER CODE END WHILE */
    if(!state&&HAL_GetTick()-lastPushTick>=200){
        HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_RESET);
    }
    if(!state&&HAL_GetTick()-lastPushTick>=1000){
        HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_SET);
        lastPushTick=HAL_GetTick();
    }

    if (HAL_UART_Receive(&huart2, buf+255, 1, 100) == HAL_OK) {
        for(int i=0;i<255;i++){buf[i]=buf[i+1];}
        if(buf[254]=='S')(HAL_GPIO_WritePin(GPIOA,GPIO_PIN_5,GPIO_PIN_SET);state=1;
        HAL_UART_Transmit(&huart2, buf+254, 1, 100);}
        if(buf[254]=='R'){ sprintf(buttoncountwrite, "%d\r\n", buttoncount);
        HAL_UART_Transmit(&huart2, buttoncountwrite, strlen(buttoncountwrite), 100);
        HAL_UART_Transmit(&huart2, buf+254, 1, 100);state=0;buttoncount=0;
        }
    }

    if (HAL_GPIO_ReadPin(GPIOC, GPIO_PIN_13) == GPIO_PIN_RESET) {
        buttoncount++;
        lastPushTick2 = HAL_GetTick();
        while (HAL_GetTick()-lastPushTick2<=100)
        while (HAL_GPIO_ReadPin(GPIOC,GPIO_PIN_13)==GPIO_PIN_RESET);
    }
}
```

Implement the following:

- 1 The indicator LEDs blinks every one second (500ms on, 500ms off) after reset
- 2 Pushing the USER button the first time will change the state such that only one LED is blinking. (choose any)
- 3 Pushing the USER button the second time will change the state such that only two LEDs are blinking.
- 4 Pushing the USER button the third time will go back to the first state after reset

```
int state=0;
int lastPushTick=0;
int lastPushTick2=0;
/* USER CODE END 2 */

/* Infinite loop */
/* USER CODE BEGIN WHILE */
while (1)
{
    if (state==0&&HAL_GetTick()-lastPushTick>=500) {
        HAL_GPIO_TogglePin(GPIOA,GPIO_PIN_5);
        HAL_GPIO_TogglePin(GPIOA,GPIO_PIN_8);
        HAL_GPIO_TogglePin(GPIOA,GPIO_PIN_9);
    }
    else if (state==1&&HAL_GetTick()-lastPushTick>=500) {
        HAL_GPIO_TogglePin(GPIOA,GPIO_PIN_5);
    }
    else if (state==1&&HAL_GetTick()-lastPushTick>=500) {
        HAL_GPIO_TogglePin(GPIOA,GPIO_PIN_5);
        HAL_GPIO_TogglePin(GPIOA,GPIO_PIN_8);
    }

    if (HAL_GPIO_ReadPin(GPIOC, GPIO_PIN_13) == GPIO_PIN_RESET) {
        state=(state+1)%2;
        lastPushTick2 = HAL_GetTick();
        while (HAL_GetTick()-lastPushTick2<=100)
        while (HAL_GPIO_ReadPin(GPIOC,GPIO_PIN_13)==GPIO_PIN_RESET);
    }
}
```

1. จงเขียนโปรแกรมที่ทำให้อุปกรณ์มีคุณสมบัติดังต่อไปนี้:

- อุปกรณ์รับข้อมูลเข้าทาง Keyboard ผ่านทาง Serial Communication (Baudrate เท่าไรก็ได้)
- เมื่อรับข้อมูลเข้าจาก Terminal ไม่ว่าเป็นตัวอักขระอะไรให้ พิมพ์ตัวอักขระคืนมา (Echo)
- เมื่อกดปุ่ม ให้พิมพ์จำนวนครั้งที่ได้กดปุ่มไปแล้ว ในแต่ละครั้ง พร้อมขึ้นบรรทัดใหม่ (ไม่มี Test case ที่กดเกิน 255 ครั้ง)

```
int state=0;
uint32_t i,j;
char c[3];
char buf[256];
uint32_t lastPushTick =0;

while (1)
{
    if (HAL_UART_Receive(&huart2, c+3, 1, 100) == HAL_OK) {
        for(int i=0;i<3;i++){
            c[i]=c[i+1];
            HAL_UART_Transmit(&huart2, c+2, 1, 100);
        }
    }

    if (HAL_GPIO_ReadPin(GPIOC, GPIO_PIN_13) == GPIO_PIN_RESET) {
        lastPushTick=HAL_GetTick();
        state=(state+1);
        sprintf(buf, "%d\r\n", state);
        // Transmitted with UART
        HAL_UART_Transmit(&huart2, buf, strlen(buf), 1000);
        while (HAL_GetTick()-lastPushTick<=100)
        while (HAL_GPIO_ReadPin(GPIOC,GPIO_PIN_13)==GPIO_PIN_RESET);
    }
}
```